Practical No.1

Objective: Write a Program to implement the Linear and Binary Search Algorithm.

Pseudo Code :

Code:

```
#include<iostream>
using namespace std;
//Lineary search
int L_search(int arr[], int n, int x, int i) {
   if(i>n) {
      return -1;
   }
   else{
      if(arr[i]==x){
         return i;

      }else{
         return L_search(arr, n,x,i+1);
      }
   }
}
//Binary Search
int B_search(int arr[],int x,int low ,int high){
   if(low>high){
      return -1;
   }
   else{
      int mid=(low+high)/2;
      if(arr[mid]==x){
         return mid;
      }
```

```
        else if(arr[mid]>x){
            return B_search(arr,x,low,mid-1);
        }
        else{
            return B_search(arr,x,mid+1,high);
        }
    }
}
int main(){
    int arr[]={1,2,3,4,5,6};
    cout<<L_search(arr,6,5,0)<<endl;
    cout<<B_search(arr,5,0,5)<<endl;
}
```

Output:



Practical No. 2

Objective: Write a program to implement Bubble Sort Algorithm.

Pseudo Code:

Code:

```
#include<iostream>
using namespace std;
int bubbleSort(int arr[],int n){
   for(int i=0;i<n-1;i++){
      for(int j=0;j<n-i-1;j++){
         if(arr[j]>arr[j+1]){
            int temp=arr[j];
            arr[j]=arr[j+1];
            arr[j+1]=temp;
         }
```

```
        }
    }
    return 0;
}
int main(){
    int arr[]={3,4,6,5,1};
    bubbleSort(arr,5);
    for(int i=0;i<5;i++){
        cout<<arr[i]<<" ";
    }
    return 0;
}
```

Output:



Practical No. 3

Objective: Write a Program to implement Insertion Sort Algorithm.

Pseudo Code:

Code:

```
#include<iostream>
using namespace std;
void InsertionSort(int arr[],int n){
   for(int j=1;j<n;j++){
      int temp=arr[j];
      int i=j-1;
      while(i>=0 && arr[i]>temp){
         arr[i+1]=arr[i];
         i=i-1;
      }
      arr[i+1]=temp;
```

```
    }
}
int main(){
    int arr[]={3,4,6,5,1};
    InsertionSort(arr,5);
    for(int i=0;i<5;i++){
        cout<<arr[i]<<" ";
    }

}
```

Output:

